

Exercises: Linear Algebra Recall (for Optimization)

1. Diagonalization and spectrum

Exercise 1 (Spectrum and invertibility). Let $A \in \mathbb{R}^{n \times n}$.

- (a) Show that A is invertible if and only if $0 \notin \sigma(A)$.
- (b) Show that if A is invertible then every eigenvalue of A^{-1} is of the form $1/\lambda$ where $\lambda \in \sigma(A)$.

Exercise 2 (Distinct eigenvalues $\Rightarrow$ diagonalizable). Assume $A \in \mathbb{R}^{n \times n}$ has n distinct real eigenvalues. Show that A is diagonalizable over $\mathbb{R}$.

Exercise 3 (Orthogonal diagonalization is special). Give an example of a diagonalizable (real) matrix A that is not orthogonally diagonalizable (i.e., there is no orthogonal Q with $A = Q\Lambda Q^\top$).

2. PSD order and spectral theorem

Exercise 4 (Loewner order is a partial order). On S_n (real symmetric matrices), define $A \preceq B$ iff $B - A \succeq 0$. Show that $\preceq$ is a partial order on S_n (reflexive, antisymmetric, transitive).

Exercise 5 (PSD/PD and eigenvalues). Let $A \in S_n$ with spectral decomposition $A = Q\Lambda Q^\top$.

- (a) Prove $A \succeq 0$ if and only if all eigenvalues satisfy $\lambda_i \geq 0$.
- (b) Prove $A \succ 0$ if and only if all eigenvalues satisfy $\lambda_i > 0$.
- (c) Assume $A \succeq 0$. Define $A^{1/2} = Q\Lambda^{1/2}Q^\top$ with $\Lambda^{1/2} = \text{diag}(\sqrt{\lambda_i})$. Show that $A^{1/2}A^{1/2} = A$.

Exercise 6 (Rayleigh quotient extrema). Let $A \in S_n$ and define $R_A(x) = \frac{x^\top Ax}{x^\top x}$ for $x \neq 0$. Show that

$$\lambda_{\min}(A) = \min_{\|x\|_2=1} x^\top Ax, \quad \lambda_{\max}(A) = \max_{\|x\|_2=1} x^\top Ax$$

Exercise 7 (Eigenvalue bounds from PSD order). Let $A, B \in S_n$ and assume $A \preceq B$. Prove that $\lambda_{\min}(A) \leq \lambda_{\min}(B)$ and $\lambda_{\max}(A) \leq \lambda_{\max}(B)$.

3. Orthogonal projectors

Exercise 8 (Projector onto a subspace given an orthonormal basis). Let $Q \in \mathbb{R}^{n \times k}$ have orthonormal columns ($Q^\top Q = I_k$). Define $P = QQ^\top$.

- (a) Prove that P is an orthogonal projector.
- (b) Prove that $\text{Im}(P) = \text{Im}(Q)$.
- (c) Show that for all x , Px is the unique minimizer of $\min_{y \in \text{Im}(Q)} \|x - y\|_2$.

Exercise 9 (Projection onto a hyperplane). Let $a \in \mathbb{R}^n$ with $a \neq 0$ and consider the hyperplane $H = \{x \in \mathbb{R}^n : a^\top x = b\}$.

- (a) Show that the Euclidean projection of x_0 onto H is

$$\Pi_H(x_0) = x_0 - \frac{a^\top x_0 - b}{\|a\|_2^2} a.$$

- (b) Deduce the projector onto the subspace $\{x : a^\top x = 0\}$.

4. Norms and inner products

Exercise 10 (Norm inequalities). Let $x \in \mathbb{R}^n$. Prove

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2 \leq n \|x\|_\infty.$$

Exercise 11 (Cauchy-Schwarz equality case). Let $\langle \cdot, \cdot \rangle$ be an inner product and $\|x\| = \sqrt{\langle x, x \rangle}$. Prove that for $x \neq 0$ and $y \neq 0$, equality in $|\langle x, y \rangle| \leq \|x\| \|y\|$ holds if and only if $y = \alpha x$ for some $\alpha \in \mathbb{R}$.

Exercise 12 ($\|\cdot\|_Q$ norm and eigenvalue bounds).

Let $Q \in S_n^+$ and define $\|x\|_Q := \sqrt{x^\top Q x}$.

- (a) Show that $\|\cdot\|_Q$ is a norm if and only if $Q \succ 0$.
 (b) Assume $Q \succ 0$. Show that

$$\|x\|_Q \leq \sqrt{\lambda_{\max}(Q)} \|x\|_2, \quad \|x\|_2 \leq \frac{1}{\sqrt{\lambda_{\min}(Q)}} \|x\|_Q.$$

5. Dual norms and operator norms

Exercise 13 (Dual norm basics). Let $\|\cdot\|$ be a norm and define its dual norm by $\|y\|_\star := \sup_{\|x\| \leq 1} y^\top x$.

- (a) Prove the generalized Cauchy–Schwarz inequality: $|y^\top x| \leq \|y\|_\star \|x\|$.
 (b) Compute the duals of $\|\cdot\|_2$, $\|\cdot\|_1$, and $\|\cdot\|_\infty$.

Exercise 14 (Induced operator norm). Given a vector norm $\|\cdot\|$, define

$$\|A\|_{op} := \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

- (a) Show that $\|A\|_{op} = \sup_{\|x\| \leq 1} \|Ax\|$.
 (b) Prove submultiplicativity: $\|AB\|_{op} \leq \|A\|_{op} \|B\|_{op}$.
 (c) For the Euclidean norm, show that $\|A\|_2^2 = \lambda_{\max}(A^\top A)$.

6. Linear systems and factorizations

Exercise 15 (LU and triangular solves). Assume $A = LU$ with L lower triangular and U upper triangular, both invertible. Show that solving $Ax = b$ reduces to two triangular solves.

Exercise 16 (Cholesky and SPD). (a) Let $A \in S_n$ and assume $A = LL^\top$ for some invertible lower triangular L . Prove that $A \succ 0$.

- (b) Compute the Cholesky factorization of $A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$.

- (c) Reuse your factorization to solve $Ax = b$ for $b = (2, 1)^\top$ and for $b = (0, 1)^\top$.

7. Least squares problems

Exercise 17 (Solve least squares via thin QR). Let $A \in \mathbb{R}^{m \times n}$ with $m \geq n$ and assume A has full column rank. Let $A = QR$ be a thin QR factorization with $Q \in \mathbb{R}^{m \times n}$, $Q^\top Q = I_n$, and $R \in \mathbb{R}^{n \times n}$ upper triangular (hence invertible).

(a) Let $P := QQ^\top$. Prove the identity

$$\|Ax - b\|_2^2 = \|Rx - Q^\top b\|_2^2 + \|(I - P)b\|_2^2 \quad \forall x \in \mathbb{R}^n.$$

- (b) Deduce that the unique minimizer of $\min_x \|Ax - b\|_2^2$ satisfies

$$Rx^\star = Q^\top b,$$

and explain why this can be solved by backward substitution.

Exercise 18 (Normal equations square the conditioning). Assume $A \in \mathbb{R}^{m \times n}$ has full column rank. Recall $\kappa_2(A) = \|A\|_2 \|A^\dagger\|_2$ (where $A^\dagger = (A^\top A)^{-1} A^\top$) and $\kappa_2(A^\top A) = \|A^\top A\|_2 \|(A^\top A)^{-1}\|_2$.

- (a) Show that $\|A^\top A\|_2 = \|A\|_2^2$.
 (b) Show that $\|(A^\top A)^{-1}\|_2 = \|A^\dagger\|_2^2$ and deduce

$$\kappa_2(A^\top A) = \kappa_2(A)^2.$$

- (c) Interpret why this suggests avoiding normal equations in finite precision.

8. SVD, conditioning, and numerical stability

Exercise 19 (SVD on a simple matrix). Let $A = \begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}$ with $\varepsilon > 0$.

- (a) Compute the singular values of A .
 (b) Compute $\kappa_2(A)$.